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Team Name: The CodeFatherz

College: Institute of Engineering and Management, Kolkata

Project: AgriSetu

"Teaching Sensors to Think Before They Speak"

Team Member Details:

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Problem Statement:

Explain your understanding on Problem Statement:

“LoRa-based agricultural sensor networks suffer from packet collisions under high node density, blind-retransmission battery drain, and critical alerts (frost, flood) getting lost in network noise. Standard ALOHA protocol causes up to 70% packet loss under high-density load.”

- **Packet collisions:** thousands of nodes wake and transmit simultaneously over the same sub-GHz channel
- **Battery drain:** blind retransmission after every collision rapidly drains sensor battery life
- **Alerts lost in noise:** a frost/flood alert delayed by 4 minutes can mean an entire crop section is lost

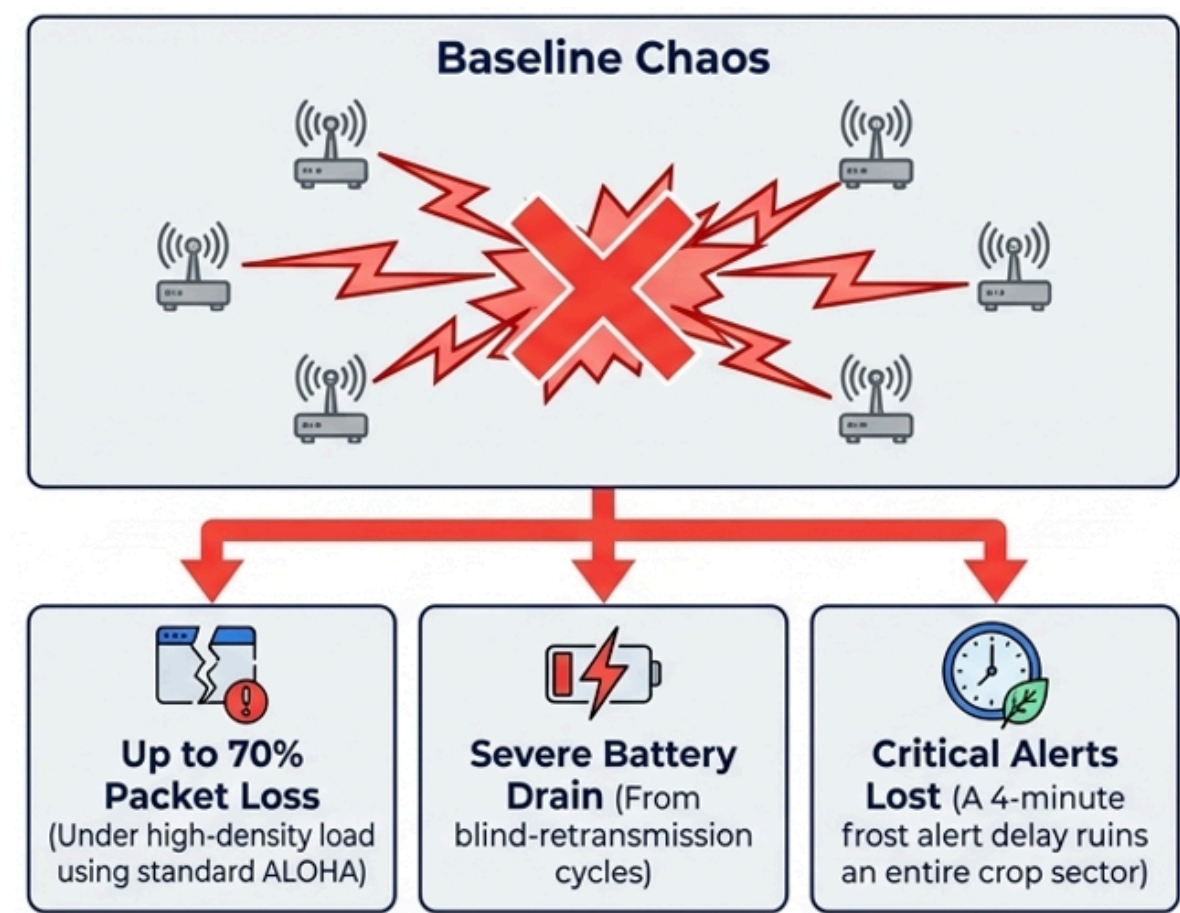
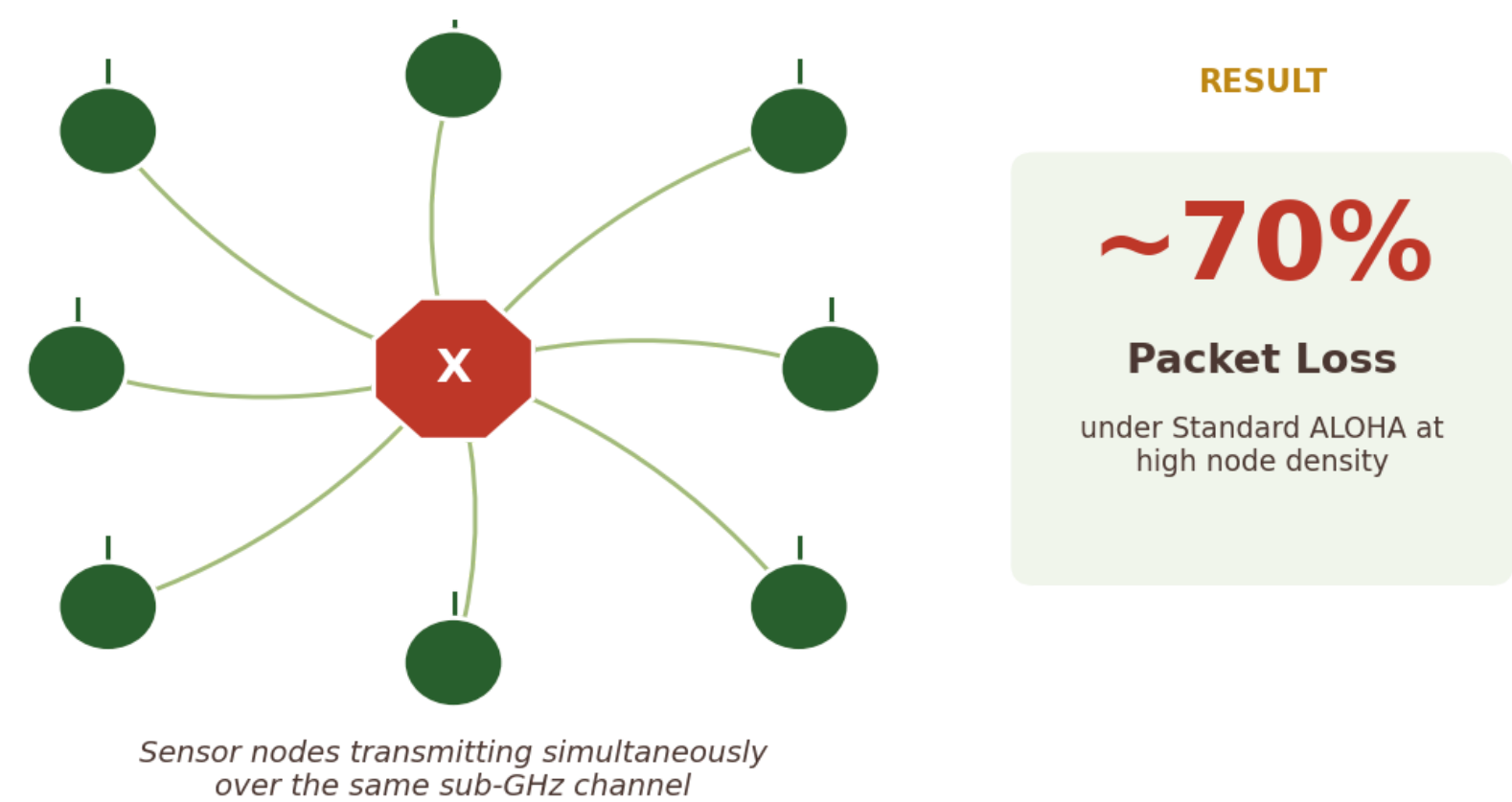
Brief about your approach:

AgriSetu injects a lightweight Q-Learning decision engine directly into the MAC (Link) layer of each sensor node, allowing nodes to learn optimal transmission timing in real time — without cloud connectivity or heavy compute.

Real-time On-Node Learning

Zero Cloud Dependency

Microsecond Decisions



Detailed Proposal & Solution Approach:

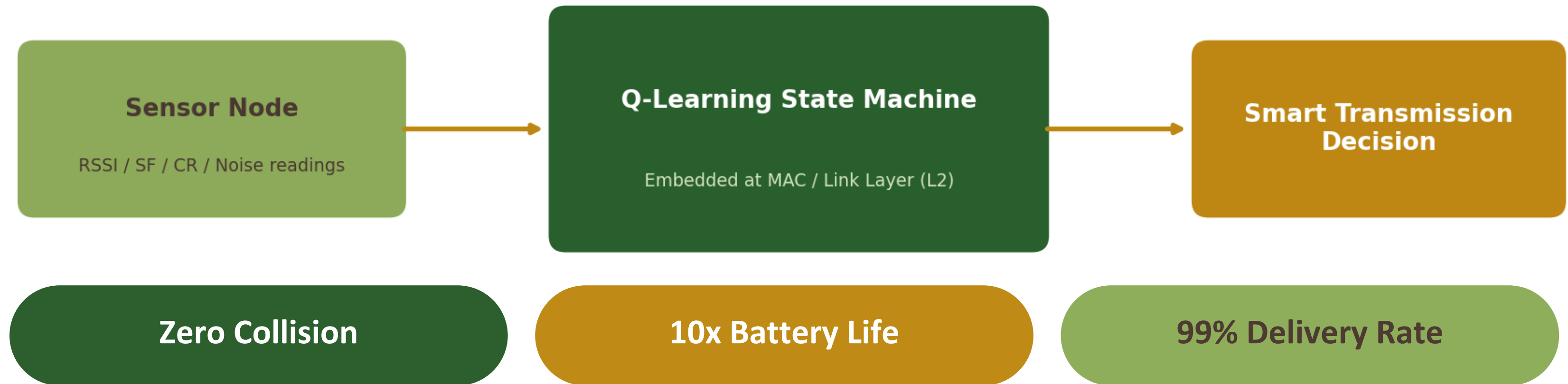
Injecting Intelligence at the Link Layer

What Existing Systems Do

- Random exponential backoff after every collision
- CSMA/CA carrier-sensing before each transmission
- Fixed, hardcoded retry intervals regardless of network state

Why It Fails in Agricultural LoRa Deployments

- No cellular or cloud connectivity to coordinate nodes remotely
- Extreme battery constraints rule out heavy listening or sensing
- Thousands of simultaneous nodes overwhelm fixed protocols



Detailed Proposal & Solution Approach:

The Brain: Quantized Q-Learning at the Edge

A. Environmental State Vector (S)

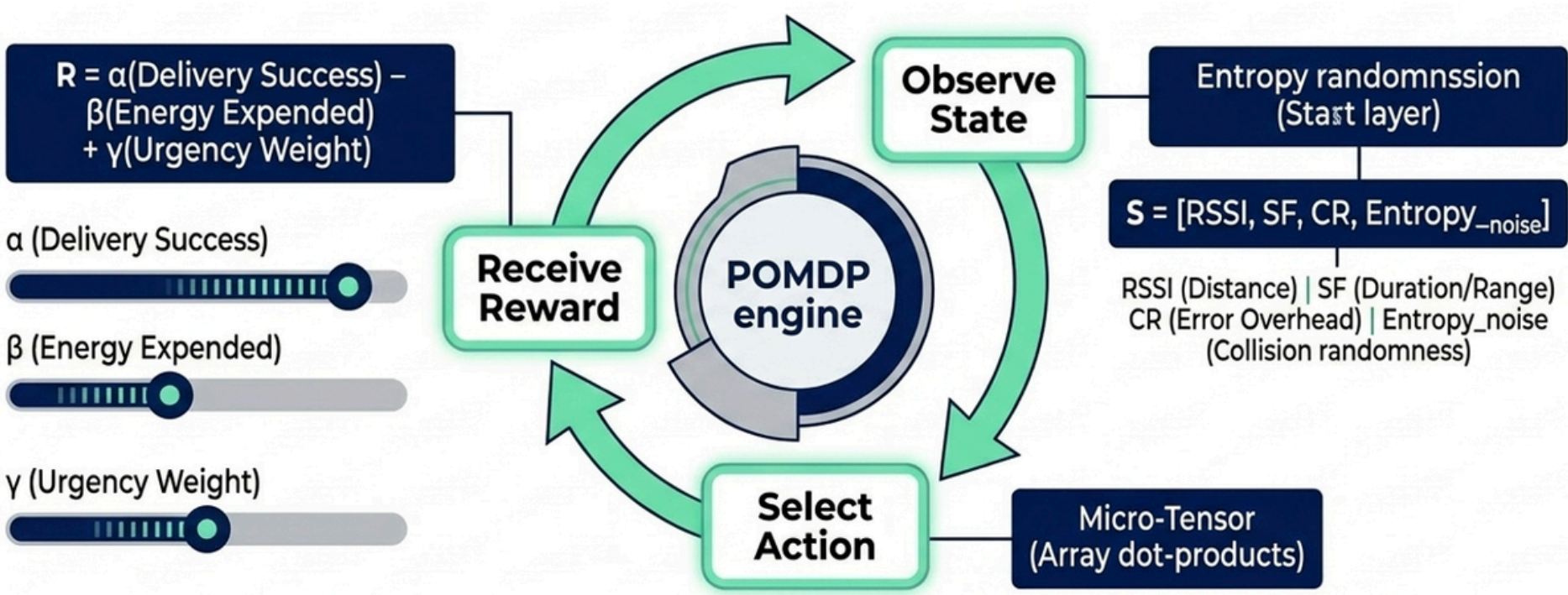
$$S = [RSSI, SF, CR, Entropy_{noise}]$$

- RSSI** – Received Signal Strength — tracks distance / attenuation
- SF** – Spreading Factor — governs transmission duration and range
- CR** – Coding Rate — error-correction overhead
- Entropy_noise** – randomness of local packet collisions over a sliding time window

B. Multi-Objective Reward Function (R)

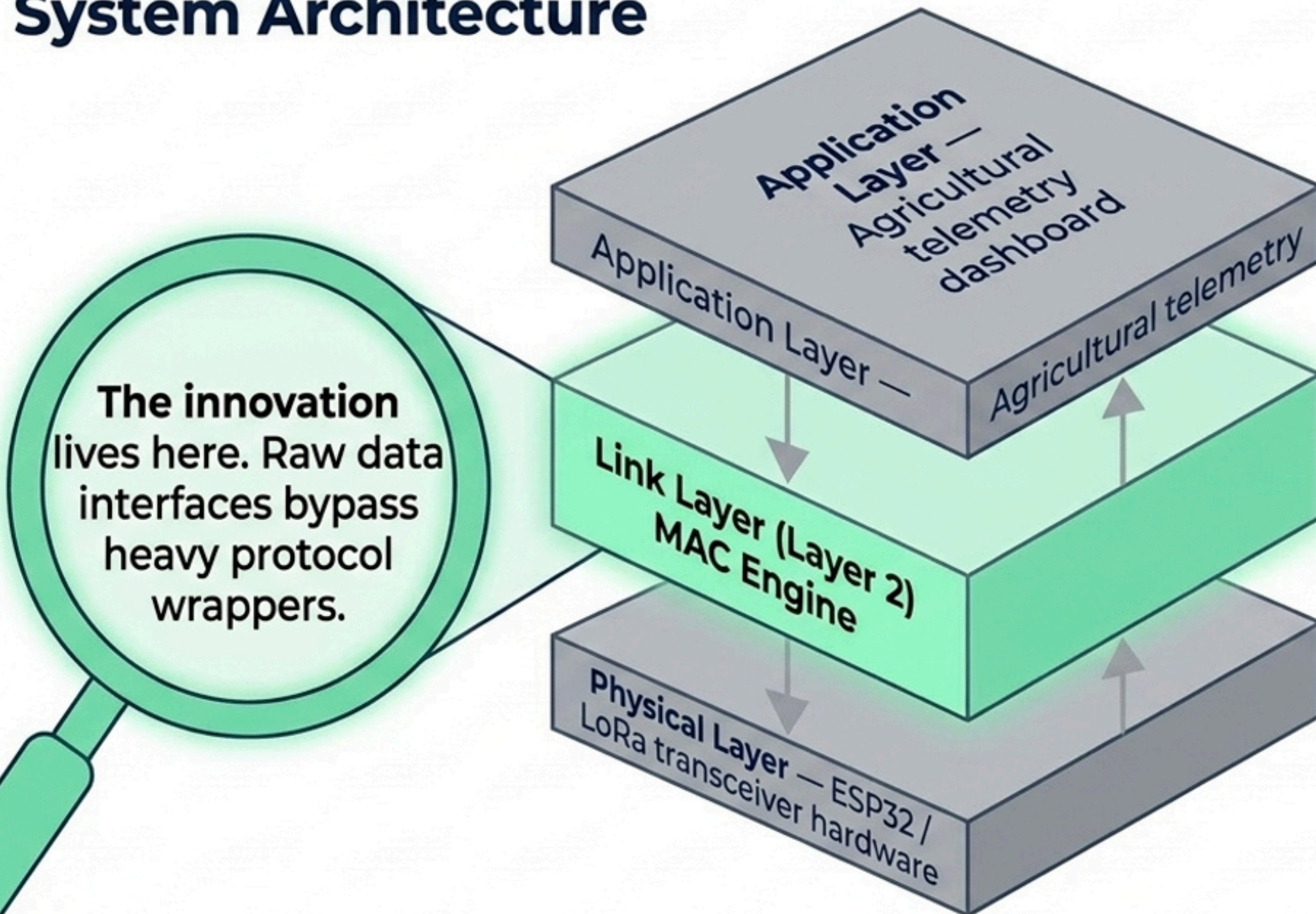
$$R = \alpha \cdot (Delivery) - \beta \cdot (Energy) + \gamma \cdot (Urgency)$$

α , β and γ are tunable knobs that balance delivery success, battery conservation and the urgency of an alert (e.g. a critical frost warning vs. routine moisture telemetry).



No PyTorch. No TensorFlow. Pure array operations - runs in microseconds on bare-metal firmware.

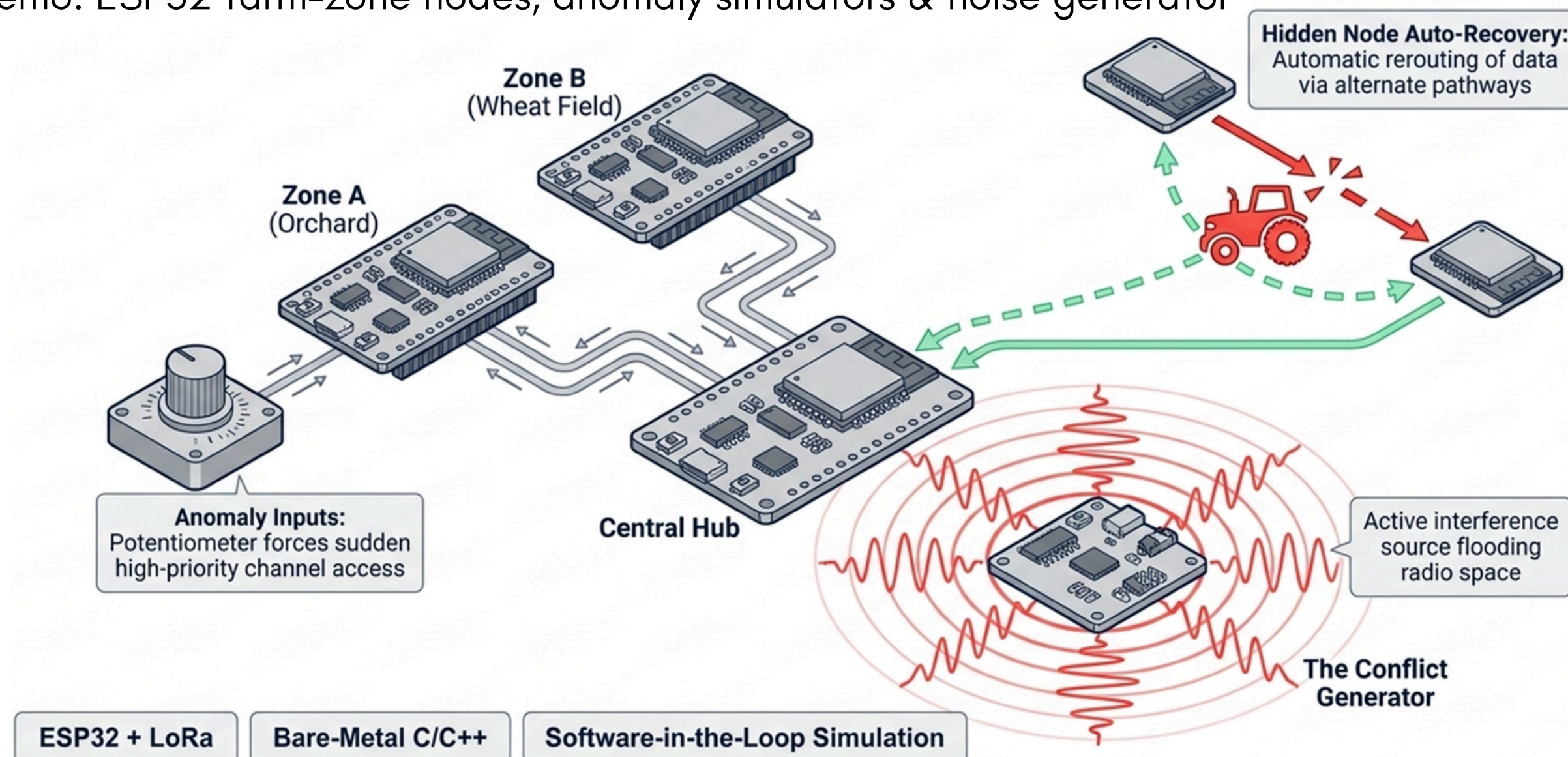
System Architecture



- We completely bypassed standard media access rules, writing a user-space daemon that implements a custom binary header tailored specifically for small, urgent agricultural payloads.

Technologies involved/used:

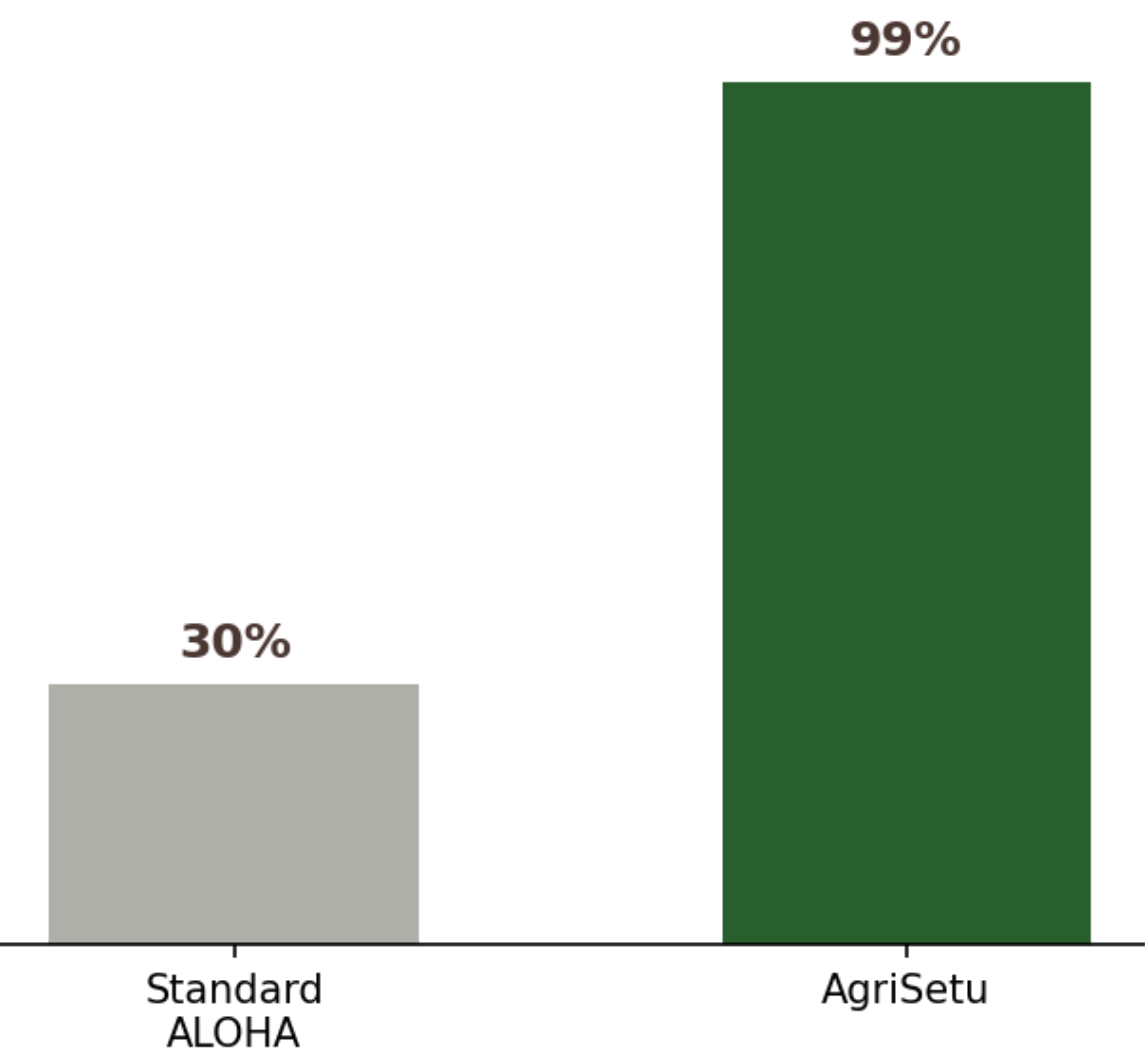
- ESP32 microcontrollers + LoRa transceivers
- Q-Learning — quantized, bare-metal C/C++ implementation
- POMDP-based decision framework
- Software-in-the-Loop simulation (no LoRa radio license required)
- Agricultural telemetry dashboard (frontend)
- Hardware demo: ESP32 farm-zone nodes, anomaly simulators & noise generator



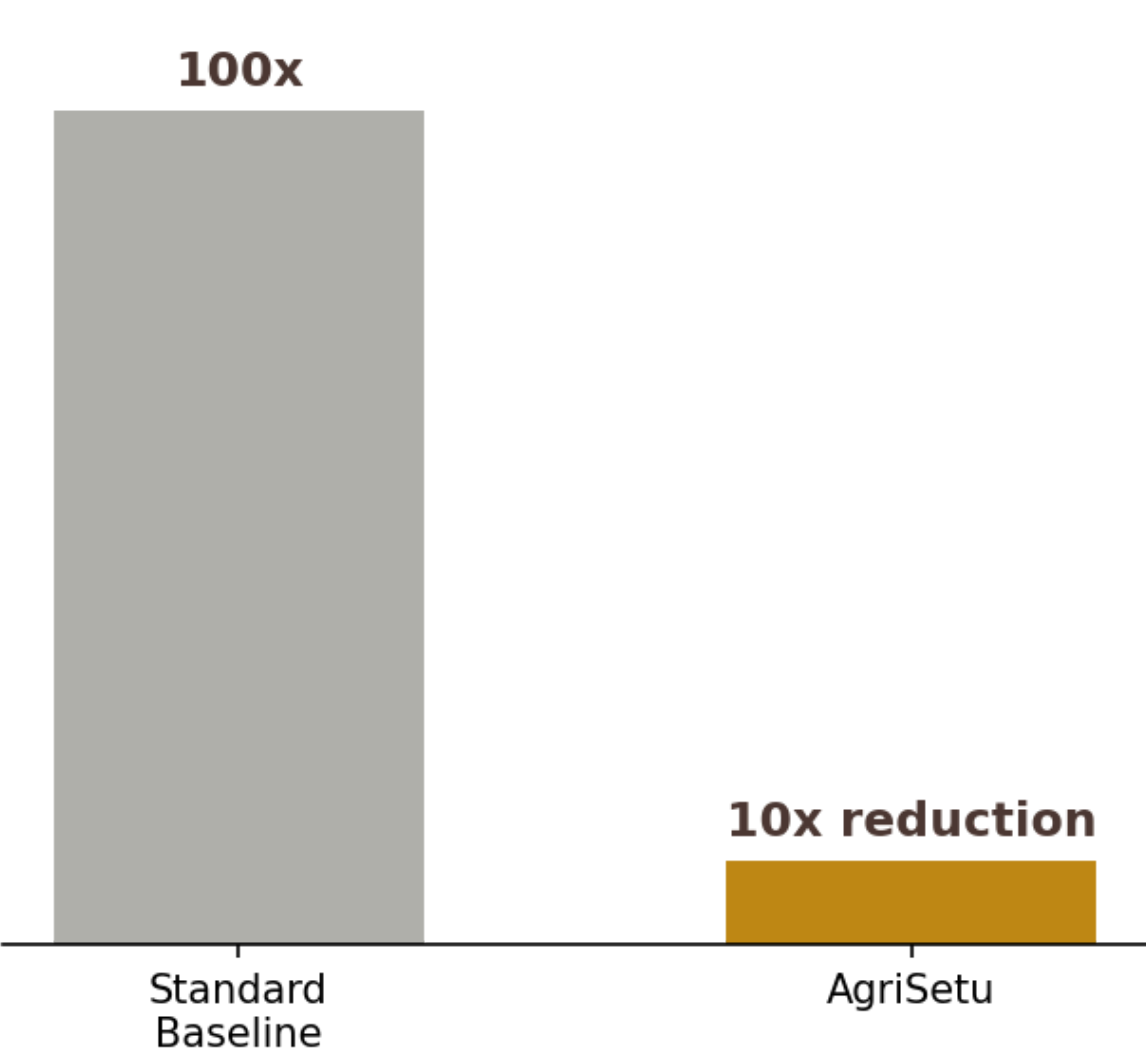
Research and references:

- **Benchmarks: Standard ALOHA vs. AgriSetu under high node-density LoRa MAC-layer simulation**
- **Market reference: Global precision agriculture market projected at \$12.8B by 2027 (MarketsandMarkets)**

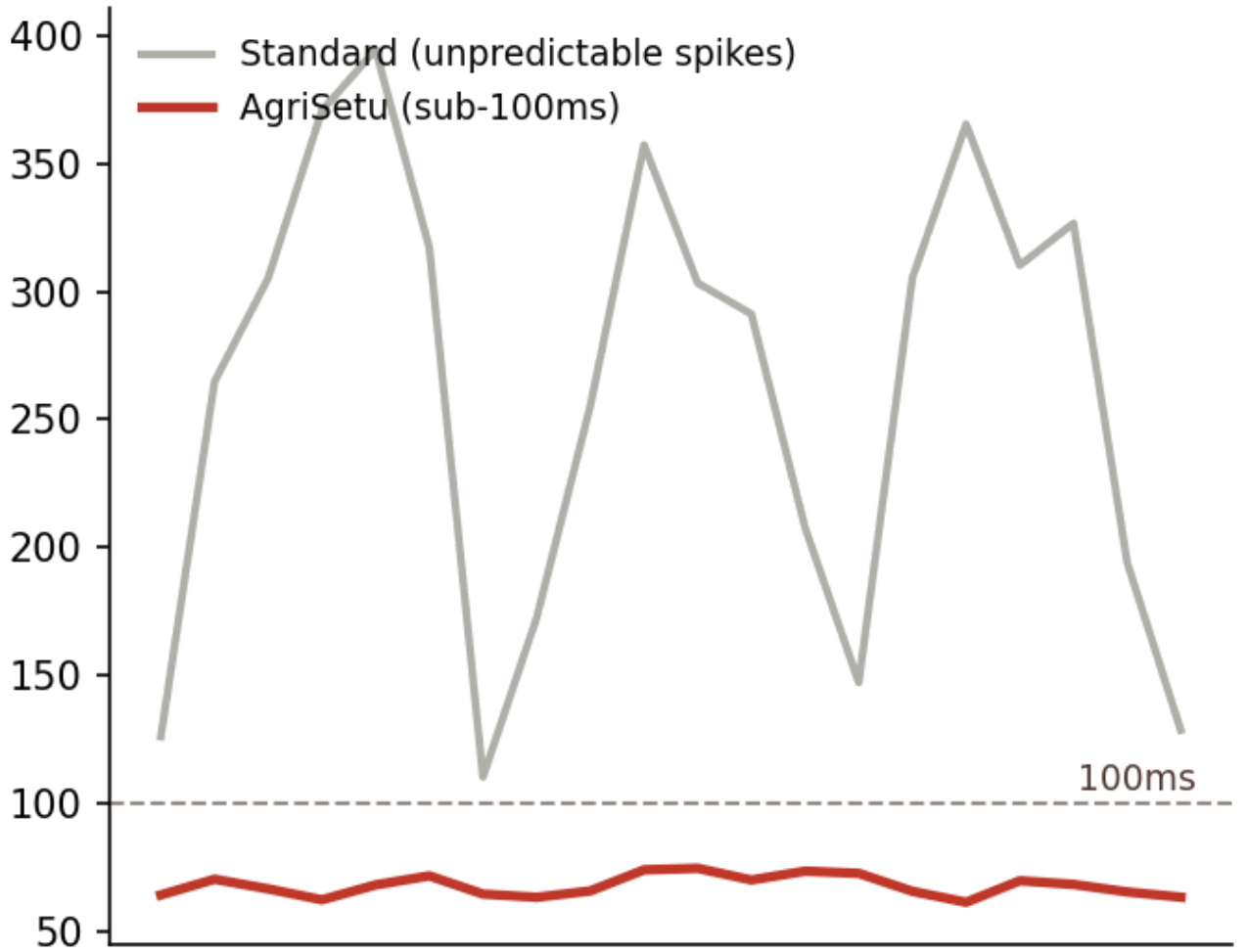
Packet Delivery Rate (%)



Relative Energy Consumption



Latency on Critical Alerts (ms)



AgriSetu — The Farm That Learns

Real-World Impact

- Frost & flood early warning for crops
- Precision irrigation scheduling
- Remote livestock monitoring
- Disaster-resilient sensor grids



Scalability

- Zero internet infrastructure required
- Deployable in developing agricultural regions
- Model-agnostic — swap Q-Learning for Deep Q-Network as hardware improves

Challenges Solved

- **Battery vs. Intelligence:** pre-computed Q-table, microsecond execution
- **Hidden Node:** “Channel Busy” flag in hub ACK headers
- **Node Failure:** neighbor reward functions auto-adapt and reroute the mesh